

FROM RELIABLE TEXT TO REAL VOICES: TRUST-AWARE PROGRESSIVE ADAPTATION FOR LOW-RESOURCE TTS

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ABSTRACT

Low-resource text-to-speech (TTS) adaptation is constrained by scarce paired data and costly manual transcription. Existing fixed-voice TTS systems can provide relatively accurate pronunciation, but their synthetic speech offers limited speaker diversity and may exhibit flat prosody. Real recordings provide natural prosody and diverse voices, yet their automatic speech recognition (ASR) pseudo-labels may contain transcription errors. We find that supervision order affects content accuracy and speaker similarity. We propose trust-aware progressive adaptation: synthetic-to-real adaptation first establishes text–speech correspondences, then restores reference-speaker control using real speech. Transcript-agreement weighting uses agreement between two fixed ASR systems as a proxy for pseudo-label reliability to limit noisy supervision. Experiments with FireRedTTS3 on Burmese and Lao and OmniVoice on Burmese show improved content accuracy with high naturalness and competitive speaker similarity. Jointly considering supervision order and pseudo-label reliability when combining synthetic and real speech offers a practical path to zero-shot voice cloning in low-resource languages with less manual transcription. Audio demos are available at <https://insiderx-pro.github.io/S2R-Adaptation-TTS/>.

Index Terms— Low-resource TTS, synthetic-to-real adaptation, pseudo-label reliability, zero-shot voice cloning

1. INTRODUCTION

Multilingual TTS enables zero-shot voice cloning from short reference recordings [1, 2, 3, 4]. Recent systems broaden acoustic-domain and language coverage [5, 6], but low-resource adaptation still lacks high-quality text–speech pairs [7, 8]. Manual transcription is costly, motivating the use of unpaired text and speech [9, 10].

These resources offer two routes to constructing training pairs. Existing TTS systems synthesize speech from text [11], providing relatively accurate pronunciation but potentially flat prosody [12]. Fixed voices also limit speaker diversity, as illustrated by the language-specific models in MMS [13]. Synthetic speech thus provides useful pronunciation supervision but limited acoustic diversity. Alternatively, ASR supplies pseudo-labels for real speech [14], preserving natural prosody and diverse speaker characteristics while introducing transcription noise.

These strengths motivate combining both sources, but synthetic augmentation can degrade speaker similarity [15]. Our controlled

comparisons show that synthetic adaptation improves content accuracy but can reduce speaker similarity; real speech restores speaker similarity, but noisy pseudo-labels can erode content accuracy. Reversing supervision order changes this trade-off, motivating joint consideration of supervision order and pseudo-label reliability.

We propose trust-aware progressive adaptation. Synthetic-to-real adaptation learns text–speech correspondences from synthetic speech before using real speech to restore reference-speaker control. Transcript-agreement weighting uses agreement between two fixed ASR systems as a proxy for pseudo-label reliability. We evaluate FireRedTTS3 [16] on Burmese and Lao and OmniVoice on Burmese.

Our contributions are:

- **Analysis.** We characterize how supervision order affects content accuracy and speaker similarity.
- **Method.** We propose synthetic-to-real adaptation with transcript-agreement weighting for noisy pseudo-labels.
- **Validation.** We test effectiveness across languages and backbones using independent ASR and controlled interventions.

2. METHOD

In Fig. 1, the Base model, Synthetic-Stage-Model, and Final-Stage-Model are abbreviated Base, S, and S→R: synthetic supervision precedes reliability-weighted real supervision.

2.1. Synthetic supervision

Fixed Teacher TTS T maps text x_i and voice v_i to $y_i^S = T(x_i, v_i)$. Filtering (Section 3) and same-voice reference pairing yield $\mathcal{D}_S = \{(x_i, y_i^S, r_i^S)\}$, with reference speech r_i^S , example index i , and S/R denoting synthetic/real data. Training uses unit weights and original input text; filtering does not guarantee correct pronunciation.

2.2. Transcript agreement

For real recording y_i^R , fixed primary ASR₁, A_1 , supplies $\tilde{x}_i = A_1(y_i^R)$; fixed secondary ASR₂, A_2 , supplies only a comparison transcript. Training uses $\mathcal{D}_R = \{(\tilde{x}_i, y_i^R, r_i^R)\}$, without transcript fusion or correction. Reference speech r_i^R is a distinct 1–10 s utterance from the same local speaker within a recording.

We measure normalized character disagreement:

$$d_i = \frac{\text{ED}(\mathcal{N}(\tilde{x}_i), \mathcal{N}(A_2(y_i^R)))}{\max(1, |\mathcal{N}(\tilde{x}_i)|)}. \quad (1)$$

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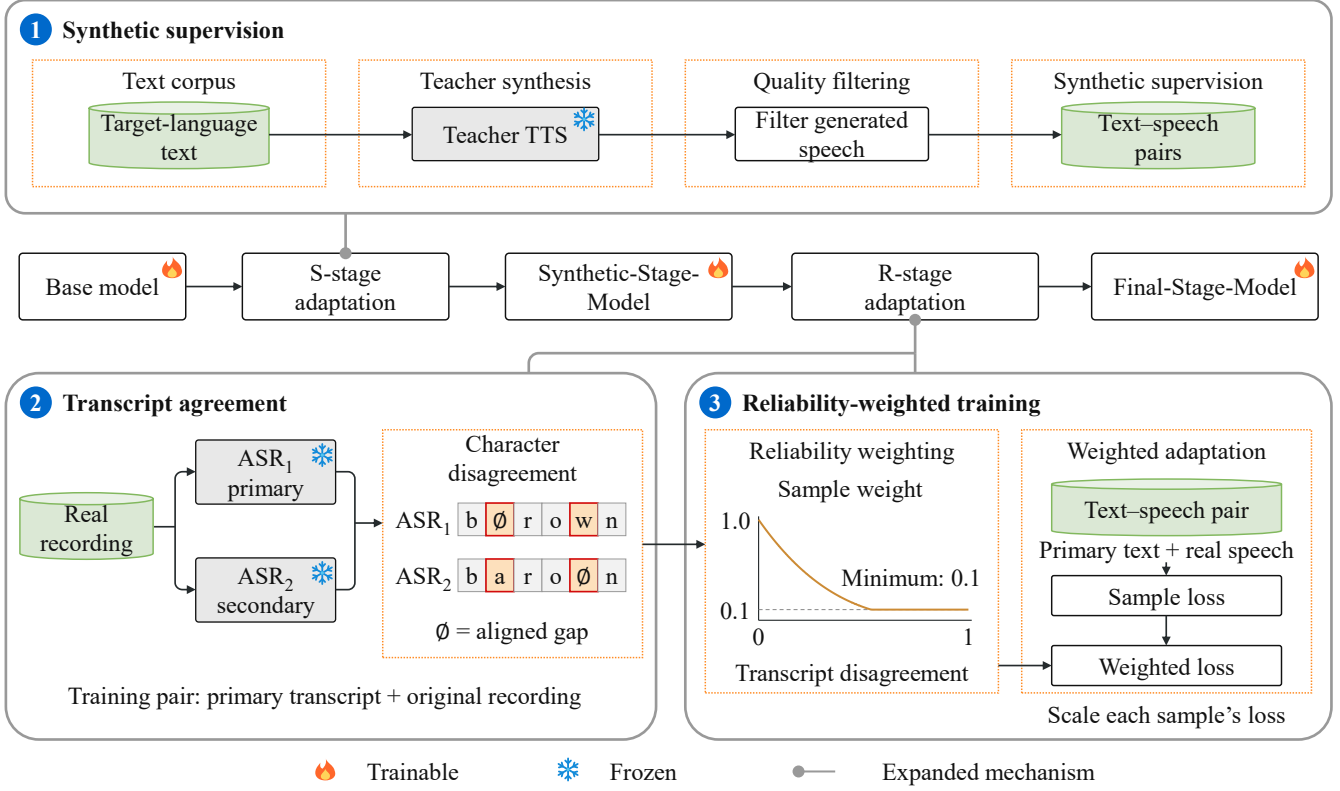


Fig. 1. Trust-aware progressive adaptation. Synthetic supervision establishes text-speech correspondences in S-stage adaptation; transcript agreement guides reliability-weighted training on real recordings in R-stage adaptation. The primary transcript remains the training text.

ED and $|\cdot|$ denote code-point Levenshtein distance and text length. Scoring-only normalization $\mathcal{N}$ applies Unicode NFC, removing whitespace and punctuation/symbol/control categories but retaining letters, digits, and combining marks, without case, numeral, or Zawygi conversion. Empty primary labels are excluded; valid empty secondary transcripts give $d_i = 1$, distinct from failed ASR requests.

Agreement does not verify correctness: recognizers may share errors. In Fig. 1, character/transcript disagreement denotes d_i ; $\emptyset$ marks an alignment gap, not a training character.

2.3. Reliability-weighted training

Reliability weighting. To limit noisy supervision’s contribution [17], we assign

$$w_i^{(\gamma)} = \max(w_{\min}, [1 - \min(d_i, 1)]^\gamma), \quad (2)$$

where $\gamma > 0$ controls decay and $w_{\min}$ floors weights. Cubic uses $\gamma = 3$, $w_{\min} = 0.10$ (Section 3): perfect agreement gets unit weight; disagreement reduces influence.

Weighted adaptation. For minibatch $\mathcal{B}$ and sample loss ℓ_θ at trainable parameters θ ,

$$\mathcal{L}_R(\theta) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} w_i^{(\gamma)} \ell_\theta(y_i^R, \tilde{x}_i, r_i^R), \quad (3)$$

averages by batch size $|\mathcal{B}|$, not weight sum. Weighting changes loss scale but need not proportionally scale AdamW updates. Offline scoring adds no inference-time ASR. Let $\mathcal{A}_B(\theta; \mathcal{D}, w)$ perform B updates from θ on data $\mathcal{D}$ weighted by w . From Base θ_0 ,

$\theta_S = \mathcal{A}_{B_S}(\theta_0; \mathcal{D}_S, \mathbf{1})$ and $\theta_{SR}^{(\gamma)} = \mathcal{A}_{B_R}(\theta_S; \mathcal{D}_R, w^{(\gamma)})$, where B_S/B_R are update budgets and $\mathbf{1}/w^{(\gamma)}$ collect unit/real-example weights. Backbones/objectives stay unchanged; stages restart optimizers/schedules. R starts from Base; R→S reverses stages.

2.4. Backbone objectives and reference conditioning

FireRedTTS3 [16] trains its core with RedAE/CAM++ [18] frozen. We weight its full flow-matching [19]/stop loss, $\ell_i = \ell_{\text{flow},i} + 0.1\ell_{\text{stop},i}$, or OmniVoice’s masked acoustic-token loss [6].

RedAE/CAM++ (omitted from Fig. 1) encode reference r as acoustic prompt $z^r = E_{\text{AE}}(r)$ and speaker embedding $e^r = E_{\text{spk}}(r)$. Controls fix pairings/reference text, exclude ineligible targets, and pair mixture references within domains. For text x and K patches [20, 21],

$$p_\theta(z_{1:K} | x, z^r, e^r) = \prod_{k=1}^K p_\theta(z_k | x, z^r, e^r, z_{<k}), \quad (4)$$

History $z_{<k}$ is ground truth in training and generated at inference; frozen encoders permit core-conditioning changes. For 16 texts, independent prompt/embedding swaps use two same-text references. For generated g , define $q = \text{SIM}(g, r_A) - \text{SIM}(g, r_B)$ (SIM-O cosines); Δ_{emb} averages the r_B -to- r_A embedding-induced change in q over prompts/texts. History tests use 8/23 prefix pairs (normalized ASR CER $\leq 5\%$), fixing reference, boundary, and subsequent random state. Suffix scoring excludes prefixes and the first 250 ms; unchanged-history replay reproduces outputs (Section 4).

Table 1. Main results. (a) CER (%), SIM-O, and H (0–100); adapted CER/SIM-O: mean $\pm$ sample SD over three seeds. (b) MOS (1–5): mean [approximate 95% crossed-bootstrap CI]. (c) Independent-ASR CER (%). Bold: numerical bests, not significance.

(a) Strategy	FireRedTTS3 / Burmese			FireRedTTS3 / Lao			OmniVoice / Burmese		
	CER $\downarrow$	SIM-O $\uparrow$	H $\uparrow$	CER $\downarrow$	SIM-O $\uparrow$	H $\uparrow$	CER $\downarrow$	SIM-O $\uparrow$	H $\uparrow$
Base	142.52	0.7244	0.00	100.45	0.7343	0.00	10.35	0.7278	80.34
S	17.43 $\pm$ 0.45	0.4513 $\pm$ 0.0041	58.36	13.49 $\pm$ 0.26	0.6108 $\pm$ 0.0006	71.61	8.21 $\pm$ 0.20	0.6864 $\pm$ 0.0010	78.54
R	54.09 $\pm$ 0.68	0.7314 $\pm$ 0.0005	56.41	37.06 $\pm$ 0.36	0.7537 $\pm$ 0.0019	68.60	10.74 $\pm$ 0.41	0.7328 $\pm$ 0.0013	80.48
R $\rightarrow$ S	14.71 $\pm$ 0.62	0.3636 $\pm$ 0.0038	50.98	10.64 $\pm$ 0.09	0.5959 $\pm$ 0.0014	71.50	7.36 $\pm$ 0.12	0.6948 $\pm$ 0.0035	79.41
S $\rightarrow$ R: 1.0	23.27 $\pm$ 0.15	0.6942 $\pm$ 0.0014	72.89	14.99 $\pm$ 0.25	0.6993 $\pm$ 0.0010	76.74	10.21 $\pm$ 0.22	0.7142 $\pm$ 0.0017	79.56
S $\rightarrow$ R: 0.5	20.91 $\pm$ 0.37	0.6996 $\pm$ 0.0005	74.24	14.60 $\pm$ 0.14	0.6968 $\pm$ 0.0026	76.74	8.03 $\pm$ 0.33	0.7094 $\pm$ 0.0009	80.10
S$\rightarrow$R: cubic	16.90 $\pm$ 0.26	0.6997 $\pm$ 0.0016	75.97	13.97 $\pm$ 0.36	0.6968 $\pm$ 0.0045	77.00	6.85 $\pm$ 0.08	0.7098 $\pm$ 0.0026	80.57

Strategy	FireRedTTS3 / Burmese		FireRedTTS3 / Lao		OmniVoice / Burmese	
	(b) MOS $\uparrow$	(c) CER $\downarrow$	(b) MOS $\uparrow$	(c) CER $\downarrow$	(b) MOS $\uparrow$	(c) CER $\downarrow$
Base	2.17 [1.97, 2.38]	103.56	2.55 [2.34, 2.76]	100.45	4.40 [4.23, 4.56]	12.62
S	3.15 [2.97, 3.34]	20.45	3.39 [3.22, 3.56]	18.03	4.23 [4.05, 4.40]	11.63
R	1.87 [1.69, 2.06]	68.64	2.12 [1.93, 2.32]	44.94	2.35 [2.15, 2.56]	12.81
R $\rightarrow$ S	2.72 [2.52, 2.92]	17.93	3.27 [3.07, 3.47]	17.13	4.32 [4.12, 4.50]	11.65
S $\rightarrow$ R: 1.0	3.78 [3.55, 4.00]	26.82	3.78 [3.61, 3.94]	23.85	4.35 [4.17, 4.51]	11.69
S $\rightarrow$ R: 0.5	3.91 [3.71, 4.11]	25.06	3.84 [3.65, 4.02]	23.33	4.42 [4.24, 4.58]	11.41
S$\rightarrow$R: cubic	4.12 [3.93, 4.30]	19.34	3.97 [3.77, 4.17]	18.53	4.51 [4.35, 4.66]	9.83

3. EXPERIMENTS

3.1. Experimental setup

Data. Burmese S/R comprise 160,918 Azure Nilar/Thiha pairs (291.148 h) and 48,043 DVB pairs (92.263 h); Lao has 119,246 S and 26,004 R pairs (200.004/45.117 h). ASR₁ is Gemini 2.5 Flash; ASR₂ is Omnilingual ASR [22]. Synthetic filtering checks valid audio/text, 0.4–20 s duration, ASR CER $\leq$ 20%, speech fraction $\geq$ 0.65, clipping $\leq$ 0.005, deduplication, evaluation exclusions, reference eligibility, and a 1,024-token limit. Burmese S starts from 171,240 candidates; 717 real targets lack references. Real speech spans 300 recordings, 1,200 local speaker groups, and 286 unverified cross-recording identity clusters.

Training. Burmese S/R use 53,640/16,015 updates at learning rates $2 \times 10^{-6}/10^{-6}$; Lao uses 39,749/8,668 updates. Orders share AdamW [23], 3% warmup and cosine decay. All adaptation/control runs use one device, accumulation/effective batch 3, and training seeds 42/17/73. Tables 1a and 3 report adapted CER/SIM-O as mean $\pm$ sample SD over these seeds; Base has no training-seed SD. Dev120 (60 texts $\times$ two references) selects $\gamma = 3$, $w_{\min} = 0.10$ by minimum three-seed mean CER with SIM-O $\geq$ 0.6970, over $\gamma \in \{0.5, 1, 2, 3, 4, 6\}$, $w_{\min} \in \{0, 0.05, 0.10, 0.20\}$. Broadcast R uses 16,015 updates, not test-selected.

Evaluation. Burmese Common400 (100 general/300 cloning conditions; 150 texts) uses Omnilingual ASR Unlimited 7B v2. Clone300 uses 30 Burmese/English/Chinese references; SIM-O averages ECAPA-TDNN/WavLM-large cosines [24, 25]. Lao uses Gemini on 404 FLEURS [26] rows (260 inputs, one reference). CER comparisons are within-language. Common400 reuses FLEURS train/development conditions. Normalized evaluation texts do not overlap archived DVB or Lao S/R transcripts. Burmese hashes match none of the 100 checked target/reference conditions. Audited Clone300 speakers are disjoint from adaptation speakers; audio near-duplicate auditing remains incomplete. FireRedTTS3 decoding uses seed 42, 10 flow steps, guidance 2.0, and stop threshold 0.5;

long/empty outputs remain scored. Joint score $H = 200as/(a + s)$ uses $a = \text{clip}(1 - \text{CER}/100, 0, 1)$, $s = \text{clip}(\text{SIM-O}, 0, 1)$, and $H = 0$ if $a + s = 0$.

Listening test. Per language, 20 listeners rated all 30 matched text-reference conditions per system (naturalness 1–5; 600 ratings/system). Burmese backbones share listeners and conditions. Approximate 95% percentile CIs use 10,000 crossed-bootstrap replicates [27], independently resampling listeners and conditions with common indices across systems within each language. We report paired cubic-minus-uniform-0.5 differences. Listening-test audio uses each adapted system’s seed-42 checkpoint.

3.2. Main results

Table 1 shows order dependence, also in matched-batch Lao: uniform S $\rightarrow$ R restores similarity but can worsen CER; R $\rightarrow$ S favors CER. Cubic has the highest observed H and mean MOS. Against uniform 0.5, paired MOS differences are $+0.210$ [0.025, 0.392], $+0.130$ [−0.047, 0.315], and $+0.090$ [−0.052, 0.230] for FireRedTTS3/Burmese, FireRedTTS3/Lao, and OmniVoice/Burmese. Only the first marginal 95% interval excludes zero; these exploratory comparisons are unadjusted for multiplicity. We do not claim consistent MOS improvement across settings. OmniVoice exceeds R in H by just 0.08/100.

Independent ASR. Dolphin-small [28] (Burmese) and XLS-R Lao [29] (Lao) rescore the same generated audio and target texts. Neither participated in pseudo-labeling, reliability estimation, filtering, or model selection. Failures, empty transcripts, and overlong outputs are not selectively excluded. Cubic improves on uniform 1.0 in all groups (Table 1c); R $\rightarrow$ S still yields lower FireRedTTS3 CER, consistent with an accuracy–similarity trade-off.

Adapted OmniVoice leads the shared-text Burmese comparison (Table 2), reducing Base CER by 3.50 points; fixed-voice SIM-O is descriptive. All outputs are scored, including 210/400 mmSpeech Tacotron cases reaching its 12.54 s limit and IMS-Toucan outputs with Burmese phoneme warnings.

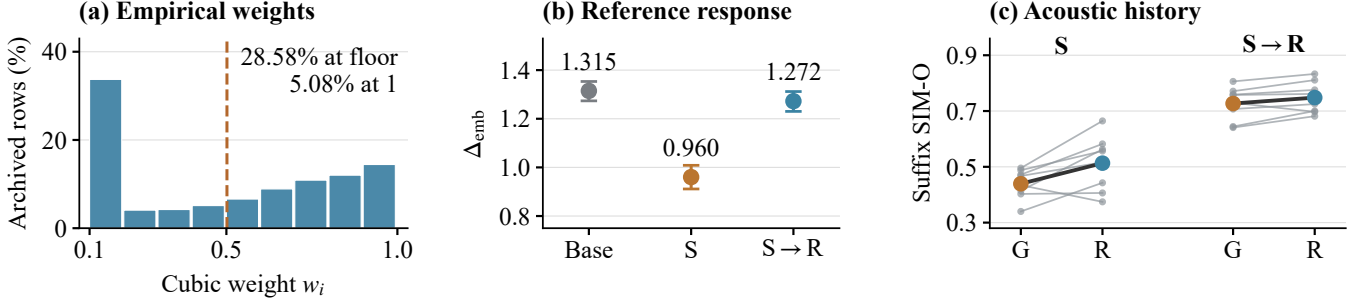


Fig. 2. (a) Cubic weights: 42,785 archived ASR rows. (b) Embedding response: 16 texts, paired-bootstrap 95% intervals. (c) Eight suffix SIM-O pairs: generated (G)/recorded (R) prefixes; thick lines: means. (b,c): historical uniform 0.5 S→R, not cubic.

Table 2. Burmese systems on Common400 / Clone300. *: fixed voices; their SIM-O scores are descriptive.

System	CER (%) ↓	SIM-O ↑
Fish Audio S2-Pro [30]	83.89	0.6376
MMS-TTS-mya* [13]	20.49	0.1029
F5-Myanmar-TTS v2 [31, 32]	35.24	0.5863
IMS-Toucan [33]	95.40	0.3298
mmSpeech Tacotron* [34]	78.70	0.0914
OmniVoice Base [6]	10.35	0.7278
FireRedTTS3 cubic	16.90	0.6997
OmniVoice cubic	6.85	0.7098

Table 3. Burmese FireRedTTS3 controls from S. Mean $\pm$ sample SD over three seeds; shuffles are averaged within seed.

Setting	CER (%) ↓	SIM-O ↑
<i>(a) FLEURS label control; S: Table 1a</i>		
Pseudo-labels, $w = 0.5$	21.17 ± 0.38	0.6345 ± 0.0024
Pseudo-labels, $w = 1.0$	23.67 ± 0.19	0.6487 ± 0.0009
Official text, $w = 1.0$	15.45 ± 0.15	0.6342 ± 0.0012
<i>(b) Broadcast weighting; uniform 1.0: Table 1a</i>		
Uniform 0.5	20.91 ± 0.37	0.6996 ± 0.0005
Linear ($\gamma = 1$)	23.25 ± 0.49	0.6956 ± 0.0030
Global shuffle	21.34 ± 0.52	0.6984 ± 0.0010
Duration-strat. shuffle	21.13 ± 0.22	0.6988 ± 0.0032
Uniform, equal mean	21.09 ± 0.55	0.6993 ± 0.0008
Cubic ($\gamma = 3$)	16.90 ± 0.26	0.6997 ± 0.0016

4. ABLATION AND ANALYSIS

4.1. Supervision order and pseudo-label reliability

Label quality. Table 3a compares labels on identical FLEURS recordings: 3,058 (12.135 h), retaining 2,921 (11.578 h) after overlap/duration exclusions. Official text reduces CER by 8.22 points at $w = 1$, retaining similarity gains; halving pseudo-label weight helps less. Label noise contributes alongside other domain effects.

Reliability validation. On these 2,921 Burmese FLEURS utterances, ASR₁ pseudo-label CER against official text, e_i , correlates with d_i : Spearman $\rho = 0.560$, Pearson $r = 0.546$. Mean label CER is 5.94% at $d_i < 0.1$ ($n = 971$) versus 32.30% at $d_i \geq 0.6$ ($n = 286$). Agreement is informative, not proof of correctness.

Weight assignment. Table 3b fixes initialization, targets, references, order, and updates. Global shuffles of cubic weights (101/202/303 crossed with three training seeds) break assignment; duration-decile shuffles control length correlations. Both preserve weights yet, like equal-mean uniform weighting, underperform cubic: assignment matters beyond loss scale. The 42,785-row archive (Fig. 2a) has mean/SD 0.50166/0.33337 and quartiles 0.10000/0.54771/0.81090. Its 588 later-excluded rows and missing later transcripts distinguish it from the 48,043-target pool (mean 0.500946). The pool’s duration-weighted mean 0.58715 becomes 0.50110/ $\approx$ 0.5865 with global/duration-stratified shuffling.

Mixtures. A 0/10/25/50/75/90/100% real-example sweep from S uses 16,015 updates with unit S/cubic R weights; synthetic replay restores less similarity than real-only continuation. Joint training from Base matches 160,920 S/48,045 R exposures and the learning-rate schedule (restart at 53,640), but yields 20.160% CER/0.65120 SIM-O, supporting progressive adaptation under matched exposure.

4.2. Reference conditioning and acoustic history

Reference response. Base/S/historical uniform 0.5 S→R yield Δ_{emb} of 1.3147/0.9602/1.2723 (Fig. 2b); matched-reference SIM-O is 0.7804/0.5672/0.7625. Conflicts follow embeddings; R restores S-weakened response; cubic is untested.

History dependence. Recorded versus generated prefixes improve suffix SIM-O by 0.0739/0.0211 for S/uniform 0.5 S→R (Fig. 2c), in 7/8 cases each. Paired 95% intervals [0.0225, 0.1204] and [0.0025, 0.0374] condition on cases, not seeds. This supports history dependence, not naturally accumulating voice drift.

5. CONCLUSION

We presented trust-aware progressive adaptation for low-resource TTS. It combines a synthetic-to-real adaptation schedule with transcript-agreement weighting. Across the evaluated languages and backbones, synthetic speech improves content accuracy but can reduce speaker similarity, while subsequent adaptation on real speech restores reference-speaker control at the risk of transcription noise. Transcript-agreement weighting helps preserve content accuracy, with improvements over uniform weighting persisting under independent ASR evaluation. Ground-truth label analysis supports transcript agreement as an informative, imperfect reliability proxy, while conditioning interventions reveal changes in reference response and sensitivity to acoustic history. Together, these findings suggest that weak supervision should be organized around the capabilities it strengthens or preserves, rather than treated as interchangeable additional data. By combining the pronunciation supervision available from existing TTS systems with the acoustic diversity of real speech, this approach offers a practical path toward zero-shot voice cloning in low-resource languages with reduced dependence on manual transcription of target-language speech.

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